UNIVERSITY OF PLYMOUTH

PUSL3190 Computing Project

Project Initiation Document (PID)

Auto parts market AI

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Chapter 01 Introduction

Background and context

The automobile sector in Sri Lanka has grown steadily over the past decade, with a large number of private vehicles being used for daily transportation. As vehicles becomes age , the demand for the spare parts increases, especially for maintenance and repairs. People who are deeply interested in this field like, owners , mechanics and small spare parts sellers are continuously involved in buying and selling automobile spare parts. Due to this high demand, the process of finding suitable spare parts remains inefficient and unreliable.

At present, most people rely on social media platforms, messaging groups, or general online marketplaces to search for spare parts. These platforms are not designed specifically for automobile spare parts, which makes the process difficult. Listings often lack proper technical details, images are unclear, and there is no structured way to verify whether a part is compatible with a specific vehicle model. This situation creates confusion and leads to frequent incorrect purchases.
From the seller's perspective, there is also no proper platform dedicated to spare parts trading. Sellers struggle to reach the correct buyers and often do not know how to price their items fairly, especially when dealing with used or rare parts. As a result, the overall spare parts market remains fragmented and inefficient.

With the advancement of artificial intelligence and web technologies, there is an opportunity to improve this situation. Intelligent systems can assist users in identifying parts, matching them with vehicles, and suggesting reasonable prices. A dedicated web-based solution can bring structure, accuracy, and reliability to the automobile spare parts market in Sri Lanka.

Problem statement

The current methods used for buying and selling automobile spare parts in Sri Lanka are inefficient and unreliable. Buyers face difficulty in identifying correct parts that match their vehicle models, which often results in wasted time and incorrect purchases. Sellers lack a dedicated platform to present detailed information about their spare parts and struggle to determine fair pricing. Existing online platforms do not provide intelligent tools such as part compatibility checking, image based identification, or price guidance. This project aims to address these problems by developing a dedicated web-based platform that uses artificial intelligence to support accurate and efficient automobile spare parts trading.

Scope and limitations

The scope of this project is the design and development of a web-based prototype system called SPAREHUBLK. The system will allow users to register, create listings for automobile spare parts, search for parts, and communicate with other users through a built-in chat system. Artificial intelligence features will be included to support text-based part matching, image based part identification, and price suggestion with a confidence indicator.
The project is limited to a prototype-level implementation and will be tested locally. It will not include online payment processing or commercial deployment. The artificial intelligence models will be trained and evaluated using limited datasets suitable for academic purposes, and their predictions will be advisory rather than guaranteed.

Expected impact and stakeholders

The expected impact of SPAREHUBLK is an improvement in the efficiency and reliability of automobile spare parts trading. Buyers will be able to find suitable parts more quickly and with greater confidence. Sellers will benefit from better visibility of their listings and guidance on pricing. Mechanics can use the platform to identify compatible parts more accurately.
Key stakeholders include vehicle owners, spare part sellers, automobile mechanics, and the project supervisor. The system also contributes academically by demonstrating the application of artificial intelligence in a real-world marketplace context.

Chapter 02 Business case

2.1 Business need

The automobile spare parts market in Sri Lanka lacks a dedicated and intelligent digital platform. Current solutions require buyers and sellers to rely on manual searching, personal knowledge, and external communication tools. This leads to inefficiencies, incorrect transactions, and dissatisfaction among users. A platform focused specifically on spare parts trading is necessary to organize listings, improve part discovery, and support informed decision-making.
The introduction of artificial intelligence into this domain can significantly enhance value by reducing search time, improving compatibility accuracy, and providing price guidance. A system like SPAREHUBLK addresses an existing market gap and improves overall transaction quality without requiring large financial investment.

2.2 Business objectives

The business objectives of this project are to improve efficiency, accuracy, and user satisfaction in automobile spare parts trading. The system aims to reduce the time required to find compatible spare parts, increase confidence in pricing decisions, and improve communication between buyers and sellers.

Chapter 04 Literature Review

Introduction and search strategy

This literature review explores research and existing systems related to online marketplaces, artificial intelligence in e-commerce, image recognition, and price prediction, with a focus on the Sri Lankan context. Sources were collected from academic databases such as Google Scholar and IEEE Xplore, as well as technical documentation and local marketplace websites. Keywords included automobile spare parts marketplace, AI-based recommendation systems, image recognition for mechanical components, and machine learning price prediction.

Online spare parts marketplaces

Online marketplaces improve convenience and accessibility for buyers and sellers. In Sri Lanka, platforms such as Ikman.lk and Riyasewana.lk are commonly used to trade vehicle spare parts. These platforms allow users to list and browse items but often lack structured information about part compatibility or accurate descriptions. This can result in mismatched purchases and longer search times for buyers.

Artificial intelligence in recommendation and matching systems

AI techniques, including machine learning and natural language processing, are widely used in e-commerce to improve product recommendations and matching. Text based matching can increase search relevance, but its effectiveness depends on the quality and consistency of listings. In Sri Lanka, many sellers provide descriptions in mixed languages or incomplete details, making accurate matching difficult. This highlights the need for intelligent systems adapted to local data conditions.

Image recognition for object identification

Convolutional neural networks have been successfully applied to image recognition tasks. In spare parts trading, image-based identification can help users who lack technical knowledge. This is particularly useful in Sri Lanka, where buyers often rely on visual inspection rather than part numbers. However, most studies are based on controlled datasets, and few examine performance in informal or second-hand market environments.

Price prediction models

Regression-based machine learning models are commonly used to predict prices from historical data and item attributes. Their accuracy is influenced by data quality and market stability. In the Sri Lankan context, spare part prices vary depending on condition, availability, and import status, and few studies focus on guidance for used parts pricing.

Identified research gap

Current literature and platforms show a lack of localized, intelligent systems for automobile spare parts trading in Sri Lanka. Existing solutions rarely combine text matching, image recognition, and price suggestion in one platform. SPAREHUBLK aims to address this gap by integrating these features into a web-based system designed for the local market and its specific challenges.

Chapter 05 Method of Approach

Research Design

This project follows a prototype-based applied research approach. The main focus is on designing, building, and testing a working system rather than only creating theoretical models. Each feature will be developed and improved step by step based on testing results and user feedback.

Data Sources and Collection

The system will use data such as spare part descriptions, images of parts, and basic vehicle information. Sample data will be collected from publicly available websites and from voluntary contributions. No personal or sensitive data will be stored. Images will be securely stored in cloud storage while testing the system to ensure data safety.

Algorithms and Tools

Text-based part matching will be done using natural language processing techniques. Image-based identification will use a convolutional neural network model implemented with TensorFlow or Keras. The price suggestion feature will use a regression based machine learning model to estimate fair prices. The frontend of the system will be built using React to provide a user-friendly interface. The backend will use Node.js to handle data and application logic, and MongoDB will store all information including user accounts, listings, and chat messages.

Evaluation Metrics and Validation

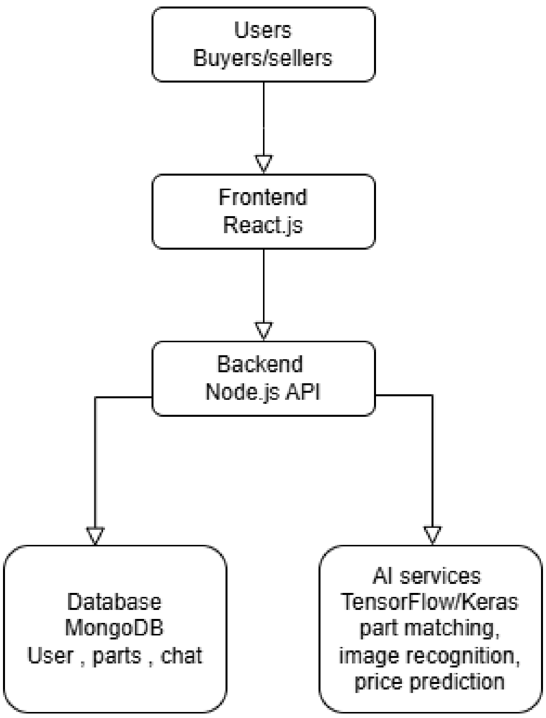
The system will be evaluated on functionality, ease of use, and the performance of AI features. Metrics will include response time, the accuracy of part matching, and feedback from users. The AI models will be tested using prepared datasets, and the results will be verified manually to ensure they are correct.

Project Management Methodology

The project will follow an incremental development approach. Each main feature will be completed, tested, and refined before starting the next. This method allows problems to be discovered early and ensures continuous improvement throughout development.

Chapter 06 Conceptual diagram

The diagram shows the main parts of SPAREHUBLK and how they interact.



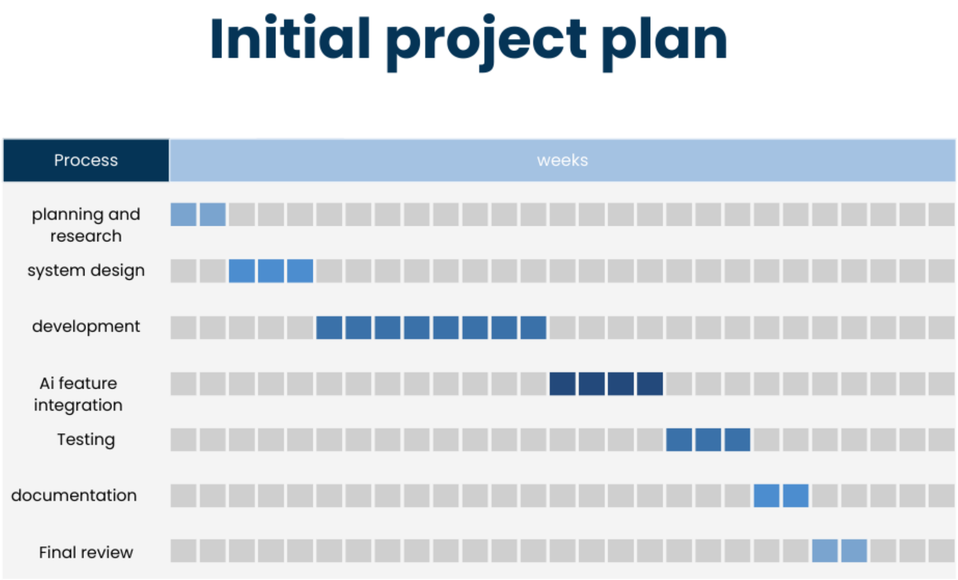
Chapter 07 Initial Project Plan

The SPAREHUBLK project is planned to be completed over seven months. The project is divided into clear phases with specific tasks and deliverables. Each phase builds on the previous one, ensuring smooth progress and continuous improvement.

Project Phases:

- Planning and Research (2 weeks) - Identify the problem, gather requirements, and review existing systems.
- System Design (3 weeks) - Create design documents, wireframes, and database schema.
- Development (8 weeks) - Build frontend, backend, and database; implement basic functionality.
- AI Feature Integration (4 weeks) - Develop and integrate AI modules for part matching, image recognition, and price suggestion.
- Testing (3 weeks) -Test system functionality, AI accuracy, and user interface usability.
- Documentation (2 weeks) - Prepare reports, diagrams, and project documentation.
- Final Review (2 weeks) - Make final adjustments and submit the project.

Gantt Chart Concept:



Chapter 08 Risk Analysis

SPAREHUBLK may face some risks that could affect its success. Identifying these risks early helps reduce problems and keep the project on track.

- A key risk is limited quality data for AI. Without enough images and descriptions, AI features like part recognition or price suggestions may not work well. To address this, we will use public datasets, augment images, and collect voluntary samples.
- Another risk is technical integration, combining frontend, backend, database, and AI modules. We will develop and test each part separately to reduce errors.
- Time management is also a concern. Delays in development or AI training could affect the schedule. Regular progress reviews and prioritizing key tasks will help manage this.
- There is a risk of low user adoption if the system is hard to use. Early testing with potential users will improve usability.
- Finally, data security and privacy are important. We will secure accounts, encrypt sensitive data, and follow privacy rules.

Key risks and mitigation chart

Key risk	Likelihood	Impact	Mitigation
Limited AI data	High	High	Use public datasets, change images to create more data, collect
Technical integration	Medium	High	Develop and test step by step, fix issues immediately
Time constraints	Medium	Medium	Stick to plan, review progress weekly, focus on important tasks
Low user adoption	Low	Medium	Test with users early, improve interface based on feedback
Security/privacy issues	Low	High	Use secure login, encrypt data, follow privacy rules

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